

Circuit description

Fig. 1 shows the block diagram of the automatic car parking space management system. The circuit (shown in Fig. 2) consists of a microcontroller (AT89C51), an LCD for displaying the number of parking spaces available, two infrared (IR) LEDs as IR transmitters (TX1 and TX2), two phototransistors (RX1 and RX2) as sensors, voltage regulator IC 7805 (IC2) and a few discrete components.

Microcontroller AT89C51 (IC1) is at the heart of the system. It is an 8-bit microcontroller with 4 kB of Flash programmable and erasable read-only memory (PEROM), 128 bytes of RAM, 32 input/output (I/O) lines, two 16-bit timers/counters, a five-vector two-level interrupt architecture, a full-duplex serial port, on-chip oscillator and clock circuitry.

Power-on reset for the microcontroller is provided by the combination of resistor R2 and capacitor C3. Switch

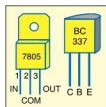


Fig. 3: Pin configurations of 7805 and BC337

S1 is used for manual reset. A 11.0592MHz crystal along with two 22pF capacitors provides the basic clock frequency to microcontroller AT89C51.

PARTS LIST

Semiconductors:

- AT89C51 microcontroller
- IC1
- IC2 - 7805, 5V regulator
- T1, T2 - L14F1 photo-transistor
- D3 - BC337 npn transistor
- T1-D5 - 1N4007 rectifier diode
- TX1, TX2 - IR LED
- LED1, LED2 - 5mm LED
- LCD - 16-character 2-line LCD

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 470-ohm
- R2, R5 - 10-kilo-ohm
- R6, R7 - 56-ohm
- R8 - 2.2-kilo-ohm
- R9 - 14-kilo-ohm
- R10 - 100-ohm
- VR1 - 10-kilo-ohm

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2 - 0.1 μ F ceramic disk
- C3 - 10 μ F, 16V electrolytic
- C4, C5 - 22pF ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 12V, 500mA secondary transformer
- S1 - Push-to-on tactile switch
- X_{cr} - 11.0592MHz crystal
- RL1 - 12V, 1 C/O relay

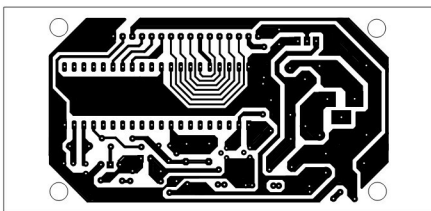


Fig. 4: An actual-size, single-side PCB for the automatic car parking space management system

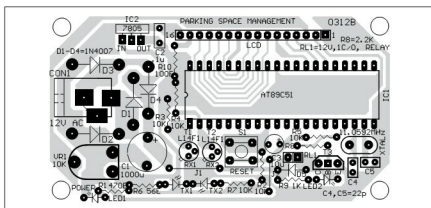


Fig. 5: Component layout for the PCB

Data pins D0 through D7 of the LCD are connected to port pins P0.0 through P0.7 of the microcontroller. The control pins of the LCD—register-select (RS), read/write (R/W) and enable (E)—are connected to port pins P2.5, P2.6 and P2.7 of the microcontroller, respectively. Preset VR1 is connected to pin 3 of the LCD for contrast control.

IR LED1 (TX1) and IR LED2 (TX2) continuously transmit infrared signal, which are received by photo-transistors T1 and T2. Photo-transistors T1 and T2 drive into saturation and so their collectors are in low states. The collectors of photo-transistors T1 and T2 are connected to the microcontroller's port pins P1.0 and P1.1, respectively.

When the infrared signal is interrupted by a car, T1 and T2 stop conducting, providing a high signal to port pins P1.0 and P1.1 of the microcontroller, respectively, indicating the presence of a vehicle between the IR transmitter-receiver pair. Port pin P3.7

of the microcontroller is used to control the barrier gate's arm. When port pin P3.7 is high, it drives transistor T3 (IC BC337) into saturation. Relay RL1 energises to lift the barrier arm up and allow the car inside the parking area.

Diode D5 connected across relay RL1 acts as a free-wheeling diode.

To derive the power supply for the circuit, the 230V AC mains is stepped down by transformer X1 to deliver a secondary output of 12V, 500mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7805. Capacitor C2 bypasses the ripples present in the regulated supply. LED1 acts as the power indicator and R1 limits the current through LED1. Fig. 3 shows pin configuration of ICs 7805 and BC337.

Construction

An actual-size, single-side PCB for the microcontroller-based automatic car